

Ankit Kumar Sahu

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SUMMARY

CSE student with a focus on systems programming and distributed systems, with hands-on experience building container runtimes, workflow engines, and low-level tools. Interested in backend and infrastructure-focused software engineering roles.

PROJECTS

- Weezy: Workflow Orchestration Engine** (*Next.js, GraphQL, BullMQ, Redis*) Github
- Built a workflow engine based on DAG execution with Redis-backed queues.
 - Set up worker processes to handle async jobs and scale execution independently.
 - Added retry + failure handling for reliable job execution.
- S-lang: Custom Programming Language** (*TypeScript*) Github
- Built an interpreter with a full pipeline: **lexer** → **parser** → **AST** → **resolver** → **interpreter**.
 - Implemented lexical scoping and variable resolution using a stack-based environment model.
 - Added support for functions, closures, and classes with dynamic typing.
 - Built REPL and file execution for running custom `.sx` programs.
- Bocker: Linux Container Runtime** (*Go*) Github
- Used Linux namespaces (PID, UTS, mount) to isolate processes inside containers.
 - Set up chroot + procfs to run commands in a sandboxed environment.
 - Applied cgroups (v1/v2) to enforce resource limits (**50 processes, 200MB memory**).
- Cax: Terminal Text Editor** (*C*) Github
- Built a terminal editor using raw mode (termios) and manual key handling.
 - Implemented syntax highlighting and efficient text rendering.

EXPERIENCE

- Open Source Contributor – Chromium** (C++, TypeScript, Java, CSS)
- Refactored Android install flow by extracting WebAPK, Add-to-Homescreen, and Universal Install handling from ChromeActivity into a new AppInstallMenuHandler module improving modularity and maintainability. CL #7458172
 - Replaced deprecated `base::Time` sentinel with `base::Time::Max()` for clearer and safer time-range handling. CL #7206813
 - Improved unreadable console output selection text color in DevTools by enhancing selection contrast for dark backgrounds. CL #7109459
- Open Source Contributor – DUB (D Package Manager)** (D)
- Auto-create missing `workingDirectory` before `dub run` execution. PR #3097
 - Fixed VisualD generator to use the selected compiler instead of hardcoded `dmd`. PR #3094
 - Corrected `ReaderException` to report actual YAML line and column. PR #3100

TECHNICAL SKILLS

- **Languages:** Go, C, C++, TypeScript, Python, Bash
- **Systems:** Linux, POSIX APIs, Process Management, Memory Management, Compilers/Interpreters, Linux Namespaces, cgroups, Docker, Basic Networking
- **Backend and Distributed Systems:** Redis, PostgreSQL, BullMQ, REST, GraphQL
- **Tools:** Git, CI/CD, AWS (EC2, S3), CMake

EDUCATION

CV Raman Global University
Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India
(2023 – 2027) — CGPA: 8.47